

DSC640 Ayden Trusty Milestone 5

March 8, 2026

1 Milestone 5 Visuals

1.1 DSC640 Ayden Trusty

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Title: Milestone 5

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Description: This program is to demonstrate my understanding/practice of using Python to review datasets and plot visuals to tell a story.

```
[1]: import pandas as pd
import matplotlib.pyplot as plt
```

```
[2]: file_path = "nationaldatabaseofchildcareprices.xlsx"
df = pd.read_excel(file_path)
```

Chart 1. Trend over time

```
[3]: trend = df.groupby("StudyYear")["MCInfant"].mean().sort_index()

plt.figure(figsize=(10, 4))
plt.plot(trend.index, trend.values, marker="o")
plt.title("Average Weekly Infant Center-Based Childcare Price Over Time")
plt.xlabel("Year")
plt.ylabel("Weekly Price ($)")
plt.tight_layout()
plt.savefig("01_trend_infant_center.png", dpi=300)
plt.show()
```

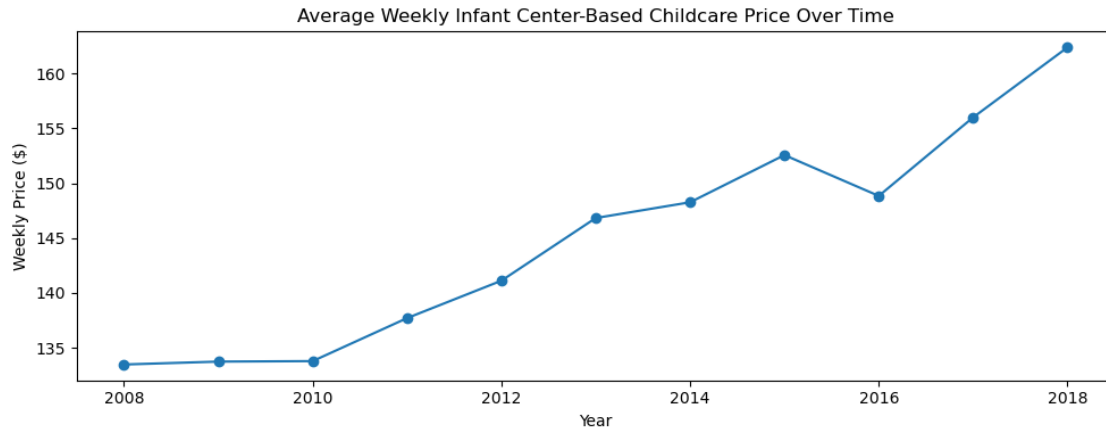
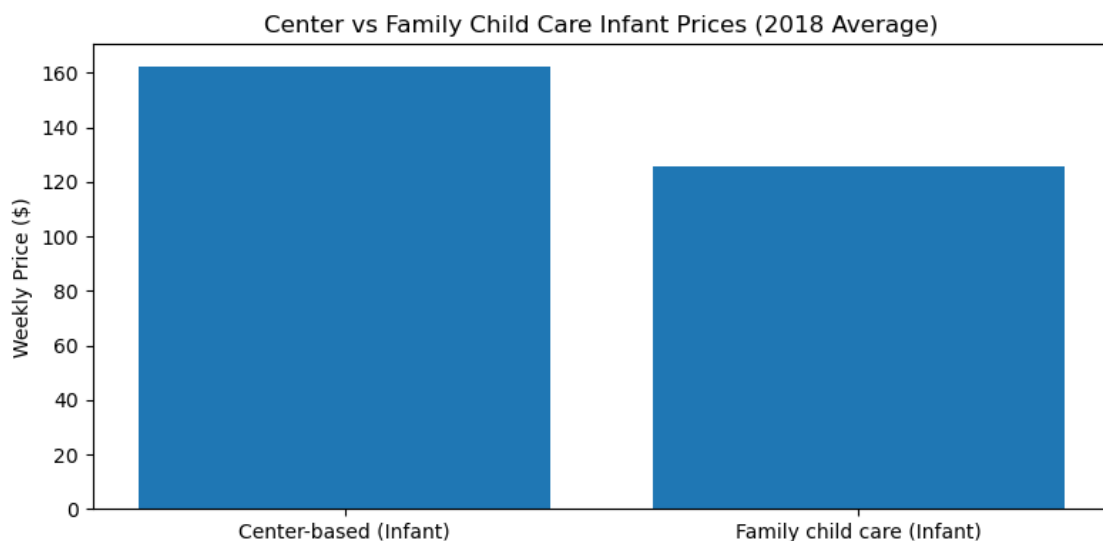


Chart 2. Care type comparison (2018)

```
[4]: df_2018 = df[df["StudyYear"] == 2018].copy()

center = df_2018["MCInfant"].mean()
fcc = df_2018["MFCCInfant"].mean()

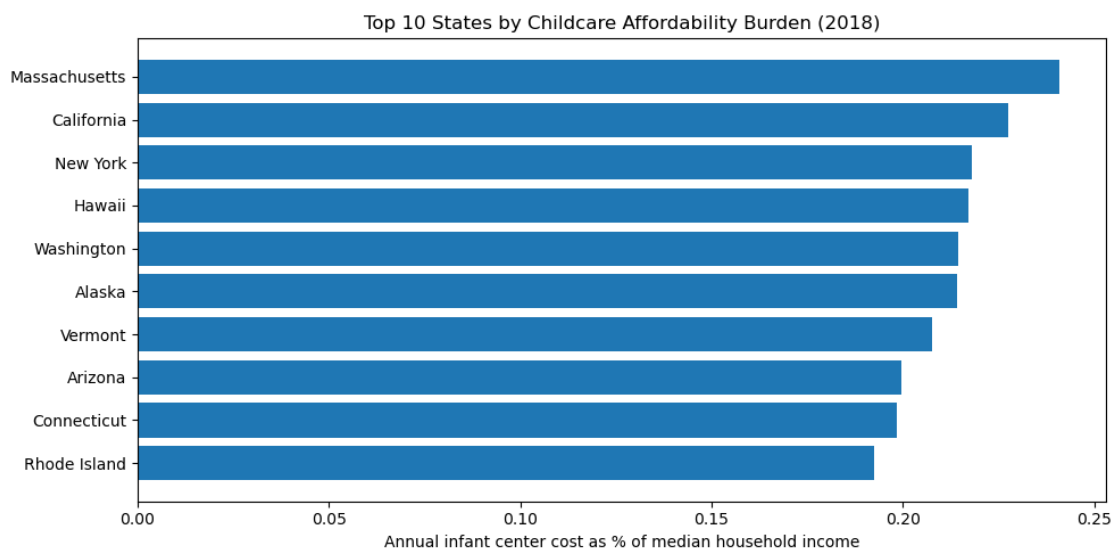
plt.figure(figsize=(8, 4))
plt.bar(["Center-based (Infant)", "Family child care (Infant)"], [center, fcc])
plt.title("Center vs Family Child Care Infant Prices (2018 Average)")
plt.ylabel("Weekly Price ($)")
plt.tight_layout()
plt.savefig("02_caretype_infant_2018.png", dpi=300)
plt.show()
```



```
[5]: df_2018 = df_2018[df_2018["MHI"].notna() & (df_2018["MHI"] > 0) &
      ↪df_2018["MCInfant"].notna()).copy()
df_2018["burden"] = (df_2018["MCInfant"] * 52) / df_2018["MHI"]

state_afford = (
    df_2018.groupby("State_Name")["burden"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure(figsize=(10, 5))
plt.barh(state_afford.index[::-1], state_afford.values[::-1])
plt.title("Top 10 States by Childcare Affordability Burden (2018)")
plt.xlabel("Annual infant center cost as % of median household income")
plt.tight_layout()
plt.savefig("03_affordability_top10_states_2018.png", dpi=300)
plt.show()
```



```
[7]: import pandas as pd
import plotly.express as px

# 1) Load only needed columns since previous attempt was extremely slow
df = pd.read_excel(
    "nationaldatabaseofchildcareprices.xlsx",
    usecols=["StudyYear", "State_Abbreviation", "MCInfant", "MHI"]
)
```

```

# 2) Keep 2018 only
df_2018 = df[df["StudyYear"] == 2018].copy()

# 3) Create affordability burden
df_2018["burden"] = (df_2018["MCInfant"] * 52) / df_2018["MHI"]

# 4) Average burden by state
state_burden = (
    df_2018.groupby("State_Abbreviation")["burden"]
    .mean()
    .reset_index()
)

# 5) Make the US heatmap
fig = px.choropleth(
    state_burden,
    locations="State_Abbreviation",
    locationmode="USA-states",
    color="burden",
    scope="usa",
    labels={"burden": "Affordability Burden"},
    title="Childcare Affordability Burden by State (2018)"
)

fig.show()

```

[]: